```
 GNU nano 8.7.1                                    exp4a.sh
#!/bin/bash

if [ -z "$1" ]; then
   echo "Usuage: $0 username"
   exit 1
fi

user_info=$(who | grep "^$1" | head-n 1)

if [ -z "$user_info" ]; then
   echo "User $1 is not currently logged in"
else
   login_time=$(echo "$user_info" | awk '{print $3 , $4}')
   start_time=$(date -d"$login-time" +%s)
  end_time=$(date +%s)
  duration=$((end_time - start_time))

   echo "User $1 logged in at: $login_time"
   echo "User $1 has been working for: $((duration/3600)) hours and $(((duration3600)/60)) minutes"
fi
```

```
vboxuser@danish:~$ nano exp4a.sh
vboxuser@danish:~$ chmod +x exp4a.sh
vboxuser@danish:~$ ./exp4a.sh vboxuser
vboxuser@danish:~$ ./exp4a.sh vboxuser
./exp4a.sh: line 8: head-n: command not found
User vboxuser is not currently logged in
vboxuser@danish:~$
```

```
  GNU nano 8.7.1
#!/bin/bash

if [ $# -ne 3 ]; then
  echo "Usage: $0 filename startline endline"
  exit 1
fi

file=$1
start_line=$2
end_line=$3

if [ -f "$file" ]; then
    sed -n "${start_line},${end_line}p" "$file"
else
   echo "file $file not found!"
exit 1
fi
```

```
  GNU nano 8.7.1
Line 1 : Welcome
Line 2 : to
Line 3 : Operating
Line 4 : Systems
Line 5 : Lab
Line 6 : Experiment
Line 7 : Done
```

```
vboxuser@Kartik:~$ touch exp4b.sh
vboxuser@Kartik:~$ nano exp4b.sh
vboxuser@Kartik:~$ nano test.txt
vboxuser@Kartik:~$ chmod +x exp4b.sh
vboxuser@Kartik:~$ ./exp4b.sh tst.txt 2 5
file tst.txt not found!
vboxuser@Kartik:~$ ./exp4b.sh test.txt 2 5
Line 2 : to
Line 3 : Operating
Line 4 : Systems
Line 5 : Lab
vboxuser@Kartik:~$
```

```
  GNU nano 8.7.1
#!/bin/bash

if [ $# -lt 2 ]; then
     echo "Usage: $0 word file1 [file2 ...]"
exit 1
fi

word = $1
shift

for file in "$@"
do
  if [ -f "$file" ]; thenn
  sed -i "/$word/d" "$file"
echo "Deleted all lines containing '$word' from $file"
else
    echo "File $file not found!"
fi
done
```

